

COURSE OUTLINE



Basic Information:		
Course Code: CSE414		
Course Title: Research Innovation & Technical Writings		
Credits: 3	Weekly: 3	Total Week: 14

Prerequisite:

N/A

Course Content (from syllabus)

This course covers the complete process of research thinking, innovation development, and professional technical writing to help students prepare a standard FYDP Phase-I report. It introduces the fundamentals of research and innovation, types of research and research models, and guides students through FYDP topic selection, defining a real-world engineering problem, and building strong motivation and objectives. Students learn how to conduct a literature review, background study, and gap analysis, and how to identify Complex Engineering Problems (CEP).

The course also covers research methodology and system design, including problem formulation, workflow/architecture diagram preparation, and selecting appropriate tools/techniques. Students study qualitative and quantitative research methods, ethical data handling, privacy considerations, data preparation, validation, and experimental planning. Later, the course focuses on result presentation, technical writing style, comparative analysis, discussion writing, and research ethics (originality, plagiarism, standards). It also includes societal and sustainability aspects, project management, and financial analysis (costing, timeline, task allocation). Finally, students learn to justify Knowledge Profile (KP), CEP and CEA, compile all report sections, and complete final drafting, presentation, and assessment according to FYDP standards.

Course Description/ Rationale

CSE 414 (Research Innovation and Technical Writing) is designed to build students' ability to think like researchers and communicate technical ideas in a professional and academic way. The course supports students in preparing the FYDP Phase-I report by guiding them step-by-step from selecting a researchable project topic to developing a complete report draft following standard format and quality.

The course is necessary because many students face difficulties in problem identification, literature review, research gap finding, methodology writing, and structured report preparation. Therefore, this course provides practical training on research models, background study, gap analysis, qualitative and quantitative techniques, ethical data handling, and documentation standards.

By completing the course, students will be able to produce clear and well-organized technical writing, justify their work using Knowledge Profile (KP), Complex Engineering Problem (CEP) and Complex Engineering Activity (CEA) frameworks, and present their FYDP Phase-I outcomes effectively through reports, quizzes, and presentations.

Course Objective:

- Develop a foundational understanding of research types, methodologies, and innovation frameworks.
- Cultivate critical thinking through evaluation of academic literature and ethical case studies.
- Equip students with the skills to design and implement both quantitative and qualitative research.
- Encourage innovative problem-solving through real-world case analysis and group activities.
- Enable effective academic reporting, citation, and visual representation of research findings.

Course Outcomes (COs) with Mappings:

After completion of this course, students will be able to:

COs	CO Statements	POs	Learning Domains	Knowledge Profile	Complex Engineering Problem	Complex Engineering Activities
CO1	Understand fundamental concepts of research methodologies and innovation frameworks	PO1	C1(Remember), C2 (Understand)	K2, K5	No	No
CO2	Design appropriate research methodologies and apply literature review insights to develop structured research solutions through group tasks and case studies.	PO3	C3 (Apply), C4 (Analyze)	K3, K4	Yes	EA1, EA2
CO3	Apply ethical principles and propose solutions that demonstrate responsible conduct and professional integrity.	PO8	C3 (Apply), A2(Respond), A3(Value)	K4, K5	Yes	EA3, EA4

Bloom's

Taxonomy

Cognitive Domain

C1: Remember

C2: Understand

C3: Apply

C4: Analyze

C5: Evaluate

C6: Create

Psychomotor Domain

P1: Imitation

P2: Manipulating

P3: Precising

P4: Articulating

P5: Naturalization

Affective Domain

A1: Receiving/ Attending

A2: Responding

A3: Valuing

A4: Organizing

A5: Characterizing

Knowledge Profile

K1: Fundamental Knowledge

K2: Mathematics

K3: Engineering

Fundamentals

K4: Specialist Knowledge

K5: Engineering Design

K6: Technology

K7: Society

K8: Research

CEP Attributes

EP1: Depth of Knowledge

EP2: Conflicting Requirements

EP3: Depth of Analysis

EP4: Familiarity of issues

EP5: Extend of Applicable

Code

EP6: Extend of Stakeholders involvements and Conflicting

Requirements

EP7: Interdependence

CEA Attributes

EA1: Range of resources

EA2: Level of interaction

EA3: Innovation

EA4: Consequences for society and the environment

EA5: Familiarity

Justification of CO-PO Mapping:

COs	POs	Justification of Mapping
CO1	PO1	Understanding research methodologies provides the fundamental engineering knowledge required by PO1.
CO2	PO3	Designing research methodologies to develop solutions is the direct application of designing solutions for engineering problems, as defined in PO3.
CO3	PO8	Applying ethical principles to propose solutions demonstrates professional ethics and responsible conduct , the core of PO8

Justification of Knowledge Profile (KP) Mapping:

COs	Knowledge Profile (K)	Justification of Mapping
CO1	K2, K5	Involves understanding research fundamentals (K2) and using modern information tools to explore methodologies (K5)
CO2	K3, K4	Focuses on designing and analyzing research solutions (K3) through investigation and synthesis of data (K4).
CO3	K4, K5	Applies investigative judgment in ethical contexts (K4) and uses professional tools for responsible practice (K5).

Justification of Complex Engineering Problem (CEP) Attributes Mapping:

COs	CEP Attributes (EP)	Justification of Mapping
CO1	No	Focuses on understanding concepts; no complex problem-solving involved
CO2	Yes	Involves designing research solutions and applying knowledge to open-ended problems.
CO3	Yes	Requires applying principles and ethics to practical, complex scenarios.

Course Delivery Plan/Lesson Delivery Plan:

Weeks	Discussion Topic and Book Reference	Teaching Learning Activities (TLA)	COs	POs	Assessment Strategies
1	Introduction to Research & Innovation; Overview of Research & Technical Writing; FYDP Phase-I Report Overview	Lecture, Interactive Discussion	CO1	PO1	Assignment, Quiz
2	Types of Research & Models; FYDP Topic Selection & Problem Context	Lecture, Case Study Discussion	CO1	PO1	Assignment, Quiz
3	Literature Review Fundamentals; Background Study; Gap Analysis; Preliminary CEP Identification	Lecture, Group Discussion	CO1, CO2	PO1, PO3	Assignment, Quiz
4	Research Methodology Concepts; System Design; Problem Formulation; Workflow/System Diagram	Lecture, Hands-on Design Exercise	CO2, CO3	PO3, PO8	Quiz
5	Qualitative & Quantitative Techniques; Ethical Data Handling (privacy, consent, integrity)	Lecture, Case Study Discussion	CO3, CO4	PO8	Midterm Exam
6	Data Preparation; Validation; Experimental Planning	Lecture, Hands-on Session	CO3	PO8	Assignment

	(metrics, train/test, evaluation plan)				
7	Designing Methodology & Diagrams; CEP, CEA & Knowledge Profile (KP) Justification	Group Discussion, Report Review Session	CO3, CO4	PO8	Presentation
8	Result Presentation & Technical Writing Style (figures, tables, expected results writing)	Lecture, Writing Practice	CO2, CO4	PO3	Assignment, Quiz
9	Comparative Analysis & Discussion Writing (comparison with literature, interpretation)	Lecture, Interactive Discussion	CO4	PO3	Assignment
10	Research Ethics & Standards; Originality/Plagiarism; Societal & Sustainability Aspects	Group Debate, Ethical Case Study	CO4	PO8	Assignment, Quiz
11	Project Management & Financial Analysis (timeline, costing, tools justification, tasks)	Lecture, Case Study Discussion	CO4	PO8	Assignment, Quiz
12	KP/CEP/CEA Justification + FYDP Phase-I Template Alignment; Future Scope/Limitations; Citations	Lecture, Report Alignment Workshop	CO4	PO8	Assignment
13	Full FYDP Final Report Drafting; Technical Editing; Academic Writing Quality	Final Review, Peer Feedback	CO2, CO4	PO3, PO8	Presentation
14	Final Assessment: Presentation + Report Submission + Mid/Final Exam	Presentation, Viva, Final Submission	CO4	PO8	Final Exam

PO Validation via Assessment:

Assessment Type	Mapped COs	Mapped POs	Justification
Quiz	CO1, CO2	PO1, PO3	Quizzes assess students' understanding of research concepts, innovation frameworks, literature review, gap analysis, and the ability to apply these ideas in structured research planning.
Assignment	CO1, CO2, CO3	PO1, PO3, PO8	Assignments evaluate students' ability to explore research methodologies, conduct background study and literature review, design structured solutions, and apply ethical principles in technical writing and research practice.
Mid-Term Exam	CO1, CO2	PO1, PO3	The mid-term exam evaluates students' conceptual understanding of research types, models, methodology design, problem formulation, literature review, and analysis of research gaps and structured solutions.
Presentation	CO2, CO3	PO3, PO8	Presentations assess students' ability to communicate research ideas, justify methodology and design choices, and

			demonstrate ethical awareness, professional integrity, and responsible academic conduct.
Final Report / FYDP Phase-I Draft	CO1, CO2, CO3	PO1, PO3, PO8	The final report assesses students' comprehensive ability to develop a research-based technical document, justify methodology using literature insights, apply ethical and professional standards, and present a complete FYDP Phase-I report in a structured and professional format.
Final Exam	CO1, CO2, CO3	PO1, PO3, PO8	The final exam evaluates students' overall understanding of research and innovation concepts, methodology development, technical writing, ethical issues, and the ability to integrate these elements into professional engineering research practice.

Assessment Pattern:

Assessment Tools	CO-wise Marks				Marks (Total=100)
	CO1	CO2	CO3		
Attendance	<i>Not Applicable</i>				
Quiz	<i>Not Applicable</i>				
Presentation					
Assignment					
Mid-Term	7	10	8		25
Final-Term		20	20		40
Total	7	30	28		65

CIE – Breakup (Theory) [60 marks]

Bloom's Criteria	Attendance	Class Test	Assignment	Presentation	Mid Exam	Total
	07	15	05	08	25	60
Remember	NA					
Understand		5	1	2	10	
Apply		5	2	3	15	
Analyze		5	2	3		
Evaluate						
Create						

SEE – Semester End Examination [40 marks]

Bloom Criteria	Score for the Test
Remember	
Understand	
Apply	20
Analyze	10
Evaluate	10
Create	

Learning Materials:

Textbook/Recommended Readings:

SL	Book Name	Author(s)	Publishers	Edition and year
1.	Research Methodology: Tools and Techniques	by Dr. Prabhat Pandey & Dr. Meenu Mishra	Pandey,Bridge Center	2015

Reference Books/Supplementary Readings:

SL	Book Name	Author(s)	Publishers	Edition and year
1	Research Design: Qualitative, Quantitative, and Mixed Methods Approaches	John W. Creswell		
2	Case Study Research: Design and Methods	Robert K. Yin		

Prepared by Module Leader Name: Dr. Md Alamgir Kabir Signature:	Approved by OBE Committee _____
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